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The effectiveness of integrating technology in EFL/ESL writing: a meta-analysis

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Abstract

Purpose – This paper aims to highlight a research on integrating technology into teaching and learning of second/foreign language writing.

Design/methodology/approach – In total, 18 empirical studies, involving a total of 1,281 second and foreign language learners, have been reviewed. These studies are selected from the following two databases: Web of Science and Google Scholar. The meta-analysis investigates how effect sizes vary depending on these moderators as follows: stage of writing, language context, learners' educational level and language proficiency level.

Findings – The findings of this meta-analysis have indicated that technology has a large effect on second/foreign language writing (d = 1.7217). These findings have also revealed that the two stages of writing, drafting and editing, have received most of the researchers' concern. In addition, high school and university learners have achieved a larger effect size of using technology in writing learning; beginner learners have achieved the smallest effect size.

Originality/value – To sum, the previous meta-analyses and reviews tried to explore the effect of computer on writing skills. However, some of them were limited to special groups (Williams and Beam, 2019) and some others analyzed very few studies (Little *et al.*, 2018). Therefore, a comprehensive analysis of the effect of implementing technology in writing skills is needed. The purpose of this study is to perform a meta-analysis of the primary studies about the integration of technology into writing skills. The primary goals of this meta-analysis were to: examine the overall effects of implementing technology in writing; synthesize the relationship between technology and a number of moderators such as stages of writing, language context, learners' target language proficiency and learners' educational level (school and university).

Keywords Multimedia, E-learning, Communication technologies, Technology-based writing, Learning writing, Foreign language writing, Second language writing, Meta-analysis

Paper type Research paper

Introduction

Writing is one of the most important skills for all human beings in the world. According to Graham *et al.* (2013), writing is important for all people in all sides of their lives; people use writing to keep in touch with their relatives, friends, colleagues, customers, etc; they can use writing in telling stories, sharing information, exchanging experiences, delivering thoughts, expressing ideas and creating imagined worlds. Besides, writing is a useful tool for learning and it is a necessary element for students' success in schools. Students use writing to transmit information; it can be used to "gather, preserve and transmit information widely." Therefore, we can say writing has a great influence on others.

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Writing is also an important skill in language learning. Due to its importance, several research studies have been conducted on how to develop learners' writing skills. Some of these research studies have dealt with the process of writing (Abbas, 2016; Al-Ghrafy, 2018; Bayat, 2014; Koh, 2017). Others have concentrated on the final product of the writing process or compared the product with the process approaches (Hasan and Akhand, 2010; Olson *et al.*, 2011; Philbin and Spirek, 1996).

Technology is fast becoming a key instrument in language learning. As learning writing is a principal component of language learning, technological devices have a crucial role in both English as a Foreign Language (EFL) and English as a Second language (ESL) writing training, especially in the past 20 years. In recent research studies, writing and technology are increasingly paired and the role of technology in improving writing is widely investigated (Alharbi, 2019; Barrot, 2020; Iwasaki *et al.*, 2019; Karami *et al.*, 2019; Kessler, 2020; Pearson *et al.*, 2005; Tam, 2012; Wihastyanang *et al.*, 2020). A considerable amount of studies have been published on the effect of computers and their different applications on the development of learners' writing abilities (Abdul Fattah, 2015; Ahmed, 2015; Ahmed, 2013; Bikowski and Vithanage, 2016; Janfaza *et al.*, 2014; Kashani *et al.*, 2013; Khalil, 2017; Khoshsima and Sayadi, 2016; Lam *et al.*, 2018).

Findings from different studies conducted to examine the effect of technology on writing were summarized in several reviews (Goldberg *et al.*, 2003; Little *et al.*, 2018; Williams and Beam, 2019). Goldberg *et al.* (2003) conducted a meta-analysis to identify the effect of computer technology on writing for K-12. The results indicated that the written work produced by students who use computers when learning to write was of greater length and higher quality. In addition, Little *et al.* (2018) conducted a meta-analysis to examine the effect of technology-based writing instruction on writing outcomes but only six studies were included. The results stated that educational technology had an impactful effect of on writing outcomes. Furthermore, Williams and Beam (2019) reviewed the effect of technology on writing but not in the form of a meta-analysis. The design of their study was in the form of a literature review. Another research review was conducted by Wollscheid *et al.* (2016), who compared studies using digital writing tools such as computers and tablets with those using traditional writing tools such as pen(cil) and paper.

To sum, the previous meta-analyses and reviews tried to explore the effect of computers on writing skills. However, some of them were limited to special groups (Williams and Beam, 2019) and some others analyzed very few studies (Little *et al.*, 2018). Therefore, a comprehensive analysis of the effect of implementing technology in writing skills is needed. The purpose of this study is to perform a meta-analysis of the primary studies about the integration of technology into writing skills. The primary goals of this meta-analysis are to:

- examine the overall effects of implementing technology in writing; and
- synthesize the relationship between technology and a number of moderators such as stages of writing, language context, learners' target language proficiency and learners' educational level (school and university).

Review of literature

Writing and language learning

Due to its importance, writing has a crucial role in language learning. People look at writing as one of the natural outputs of the learning process. Ariana (2010) assures that writing is one of the "central pillars of language learning." All people that have a relation with language learning should pay more attention and show a major interest in the skill of writing (Ariana, 2010, p. 134).

In recent years, writing has been considered to have a major role in promoting second language development. According to Williams (2012, p. 321), this role can be represented in three features of writing that will promote language acquisition as follows: its slower pace, the record that it leaves and the greater precision that it demands.

Technology and writing

Recently, researchers have shown an increased interest in using computer-assisted language learning (CALL) and mobile-assisted language learning (MALL) in improving learners' writing skills. Several studies have been conducted on using CALL in writing development (AbuSeileek and Abualsha'r, 2014; El-Ghonaimy, 2015; Fenglong, 2015; Ghafoori *et al.*, 2016; Jafarian *et al.*, 2012; Pirasteh, 2014; Zaini and Mazdayasna, 2015). A number of other researchers have sought to determine the effect of MALL on ESL, as well as EFL learners' writing skills (Al-Wasy and Mahdi, 2016; Hamad, 2017; Justina, 2016; Khalil, 2017; Kundu and Bej, 2019; Susanti and Tarmuji, 2016).

Besides, especially in the past two decades, a considerable number of researchers and writing instructors started to integrate technology with their writing classes, claiming that this integration would be so beneficial to the learners' writing skills. Using mobiles and their applications in writing classes, for example, will enhance the learners' motivation (Estarki and Bazyar, 2016), encourage learners to make their written pieces more creative (Kanala *et al.*, 2013), and develop learners' abilities to generate new ideas and correct their punctuation, as well as sentence structure errors (Abdul Fattah, 2015). Tam (2012) added that mobile phones give the learners an opportunity for writing practice, even outside their schools and universities.

More recent attention has focused on the benefits of using other technological devices in improving learners' writing proficiency. Iwasaki *et al.* (2019) indicate the importance of e-learning in the development of academic writing. Ellison and Drew (2020) assure the effectiveness of computer games in developing writers' thinking abilities and improving their organizational skills. A number of other authors have reported the importance of technology in building a better learning atmosphere, motivating learners to write and improving the quality and quantity of their products, saving learners' and instructors' time and decreasing the learners' errors by providing them with more editing facilities (Alghasab, 2020; Baker and Lastrapes, 2019; Gharehblagh and Nasri, 2020; Jafarian *et al.*, 2012).

Factors affecting the use of technology in writing

Stages of writing. As writing is a process, it should go through a number of stages. The classification of these stages differs from one person to another: Morgan, Hessler and Konrad, for example, reported five stages in the process of writing, namely, prewriting, drafting, revising, editing and publishing (Morgan *et al.*, 2007). On the other hand, Lloyd (2007) insisted on the importance of the planning technique at the beginning of the writing process. Another classification was done by Haluska (2006) who proposed four distinct steps, namely, thinking, planning, writing and revising. Manning (2000) claimed that the process of writing has four main stages, namely, selecting a topic, completing a draft, revising a draft and publishing. In this meta-analysis, the researcher will consider the most common four stages, namely, prewriting, drafting, editing and publishing. The prewriting stage refers to all the activities that can be done before starting to write. In the drafting stage, the writer should start writing about the topic. The editing stage is the stage in which the writer revises what he writes in the second stage. The publishing stage is the last stage where the writer submits his final draft.

Technology can be used in teaching writing and it is applicable to all of the above stages. Previous literature examined the impact of technology on these stages. Some research studies dealt with all the stages of writing (Ahmed, 2013; Janfaza *et al.*, 2014; Kashani *et al.*, 2013; Khalil, 2017). Some other research studies were restricted to two stages, namely, drafting and editing (Abdul Fattah, 2015; El-Ghonaimy, 2015; Khaksefidi, 2016; Khoshsima and Sayadi, 2016; Lam *et al.*, 2018; Malekzadeh and Najmi, 2015). As it is the main stage of writing, a lot of research studies tackled with drafting (Ahmed, 2015; Bikowski and Vithanage, 2016; Estarki and Bazyar, 2016; Ghafoori *et al.*, 2016; Hsu and Lo, 2018; Jafarian *et al.*, 2012; Kimmons *et al.*, 2017). Few research studies dealt with the stage of editing (Ciftci and Kocoglu, 2012). The studies presented, thus, far provide evidence that technology has an effective role in developing all the stages of the writing process.

Context. A large number of studies described the role of different technological devices on the writing performance of both second and foreign language learners. In the second language context, some researchers aimed to investigate the effect of web-based collaborative writing tasks on the writing scores of second language writers (Bikowski and Vithanage, 2016); other researchers seek to determine the effect of technological tools such as Chrome books on essay composition (Kimmons *et al.*, 2017); others tried to test the effect of blended learning on the writing skills of second language learners (Lam *et al.*, 2018). Together, these studies outline that these technological devices are very effective in improving the writing skills of the second language learners.

In the foreign language context, more studies were conducted to measure the effect of technology on foreign language learners' writing skills. These studies differed from each other in terms of the technological devices used. Some of these studies used the WhatsApp (Abdul Fattah, 2015; Arifani *et al.*, 2020; Khalil, 2017); other studies used the mobile phones and their applications to enhance learners' motivations and develop their abilities to write (Estarki and Bazyar, 2016; Gharehblagh and Nasri, 2020; Kanala *et al.*, 2013; Khaksefidi, 2016; Malekzadeh and Najmi, 2015); others used computer programs, games or applications (Ahmed, 2013; Alghasab, 2020; Baker and Lastrapes, 2019; Awada *et al.*, 2019; Ciftci and Kocoglu, 2012; El-Ghonaimy, 2015; Ellison and Drew, 2020; Ghafoori *et al.*, 2016; Hsu and Lo, 2018; Jafarian *et al.*, 2012; Janfaza *et al.*, 2014; Kashani *et al.*, 2013; Khoshsima and Sayadi, 2016). Overall, these studies highlight the positive effect of technology on foreign language learners' writing skills.

Language proficiency level. Previous studies investigating the effect of technology on the learners' writing performance were conducted at different learners' language levels. Some of these studies were conducted for advanced learners (Abdul Fattah, 2015; Bikowski and Vithanage, 2016; Jafarian *et al.*, 2012; Kashani *et al.*, 2013; Khalil, 2017); other studies were directed to intermediate learners (Estarki and Bazyar, 2016; Ghafoori *et al.*, 2016; Janfaza *et al.*, 2014; Khoshsima and Sayadi, 2016; Malekzadeh and Najmi, 2015); others addressed the beginners (Hsu and Lo, 2018; Kimmons *et al.*, 2017). Together these studies provide important insights into the effectiveness of technology on the writing skills of learners at all language levels, in general.

Educational level. A large and growing body of literature investigated the effect of technology on the learners' writing at different educational levels. Most of these studies were directed to university learners (Abdul Fattah, 2015; Ahmed, 2015; Bikowski and Vithanage, 2016; Ciftci and Kocoglu, 2012; El-Ghonaimy, 2015; Ghafoori *et al.*, 2016; Hsu and Lo, 2018; Khaksefidi, 2016; Khalil, 2017; Khoshsima and Sayadi, 2016); some of them are tackling with high school learners (Ahmed, 2013; Jafarian *et al.*, 2012; Kimmons *et al.*, 2017; Lam *et al.*, 2018); and some others with language institutes learners (Estarki and Bazyar, 2016; Janfaza *et al.*, 2014; Malekzadeh and Najmi, 2015). Overall, there seems to be some evidence to indicate that technology has a positive effect on the learners' writing skills in all the educational levels they might be in, with stronger effect at the university level.

Technology and writing skills: previous reviews

A limited number of reviews have dealt with the impact of technology on writing skills (Goldberg *et al.*, 2003; Little *et al.*, 2018; Williams and Beam, 2019; Wollscheid *et al.*, 2016). The findings of these studies indicated that technology-based writing instruction "yielded improvements in students" composing processes and writing skills, as well as their knowledge and use of new literacies' (Williams and Beam, 2019). They have also revealed that technology causes the writing process to be "more collaborative, iterative and social in computer classrooms as compared with paper-and-pencil environments" (Goldberg *et al.*, 2003).

In spite of the availability of these studies, the current study remains different for several reasons. First, one of these studies (Goldberg *et al.*, 2003) covered the duration from (1992–2002). Technological devices have been improved, for example, Twitter and WhatsApp were not known to people of that period. The second reason is that unlike the current study, two other studies (Williams and Beam, 2019; Wollscheid *et al.*, 2016) are research reviews; they are not conducted in the form of a meta-analysis. Regarding the fourth study (Little *et al.*, 2018), the researchers coded a limited number of studies (only six), which made the generalizability of the results difficult for us. The final reason that makes the current study differs from other reviews lies in the different moderators considered for each study. These different moderators from one study to another make some variation and cause the current study to be unique.

Recommendations of some of these studies also suggested conducting "more research to determine the exact mechanisms through which technology may impact writing skills" (Little *et al.*, 2018). Therefore, a comprehensive analysis of the effect of integrating technology in second/foreign language writing learning is required. The purpose of this study is to conduct a meta-analysis of the primary studies that investigated the effect of any of the technological devices on second/foreign language writing learning. The current study aims to achieve the following goals:

- To examine the overall effects of integrating technology in learning second/foreign language writing.
- To identify the relationship between technology and a number of moderators such as writing stages, language contexts, learners' target language proficiency and learners' educational levels.

In other words, this meta-analysis aims to answer the following two questions:

Q1. What is the effect of integrating technological devices into writing learning for second/foreign language learners?

Q2. What is the relationship between technology-based writing (TBW) and the following moderators: stage of writing, language context, learners' target language proficiency and learners' educational level?

Methods

This study follows a meta-analysis design. The goal of a meta-analysis is to understand the results of any study in the context of all the other studies (Borenstein *et al.*, 2009). In language learning, meta-analysis has been applied to support evidence-based practice. It has been used to assess the improvement over a period of time. Meta-analyses in language learning interventions have been used to compare the results obtained from experimental or quasi-experimental studies to find out the effect of the tools used.

Search strategy

To identify the potentially appropriate articles for this meta-analysis, the author conducted a search. In this search, articles were accessed through the following databases: Web of Science and the non-library database, Google Scholar. The search also covered the major journals of writing and technology such as the *Journal of Second Language Writing and Computers and Composition*. To search these databases, the following two sets and combinations of keywords were used: *CALL and Writing* and *Technology and Writing*.

Inclusion and exclusion criteria

The result of the literature searches yielded 8,858 journal articles. To find the most appropriate studies that correspond with the questions and method of this research, the titles and abstracts of the located articles were carefully screened by the author. Most of the previous meta-analysis studies have based their criteria of selection on some or all of the following: the targeted population, year of publication, study design, journal type, writing outcomes, the suitability of data for meta-analysis statistics and the topic which lies in the impact of technology or one of its components on the writing skills of the targeted population (Goldberg *et al.*, 2003; Little *et al.*, 2018; Williams and Beam, 2019; Wollscheid *et al.*, 2016).

To have more valuable results, the author of this meta-analysis does not want to be confined to a particular targeted population. Therefore, the learners' educational level is added as one of the variables in this meta-analysis. Besides, there is no time limit in this meta-analysis, in spite of the concentration on the most recent studies.

Therefore, the author considered only three inclusion criteria. First, studies included should necessarily examine the effect of any technological device on EFL/ESL writing skills. Studies that used other means for developing the EFL writing skills were excluded. Second, only studies that followed the experimental or quasi-experimental design were included; qualitative, descriptive and correlative studies were excluded. Third, only studies that presented clear data about the number of participants in each group, means and standard deviations were included. Studies that presented inadequate data in any of the above elements were excluded. After implementing these criteria, 18 studies that met the inclusion and exclusion criteria were included in this meta-analysis. These studies were marked with an asterisk (*) in the reference list. More details about the selected studies are included in Appendix 2.

Coding procedures

The studies included in this meta-analysis were coded for several variables. Some of these variables gave merely descriptive details about these studies, for example, the author's name, the journal title and the publication date. Other variables described the procedures followed in these studies, such as the technological device used (*a computer program – a mobile program – Twitter – email – virtual classrooms – wikis – Chromebook – WhatsApp*) the targeted writing stage *(prewriting – drafting – revising – editing)* and the language context of the study *(second or foreign language context)*. Some other variables described the subjects of these studies such as their ages, educational levels (*high school – institute – university – post-graduate*) and their linguistic level (*beginners – intermediate – advanced)*. The characteristics for coding the study samples are shown in Appendix 1.

Calculating effect size

To reveal the effectiveness of technology on ESL/EFL writing, the current meta-analysis calculated Cohen's effect size *d*. As recommended by Borenstein, *et al.* (2009), the effect size of each article was first calculated. This calculation was done by using the descriptive statistics of the control and experimental groups, including the mean scores *(M)*, standard deviations *(SD)* and sample sizes *(N)*. To calculate the standardized mean difference for each study, the difference between the two means (x1 and $x2$) was calculated and then divided by its pooled standard deviation. Such a standardized mean difference was used to represent each study and then compared with other studies. (The means and standard deviation of the included studies are shown in Appendix 3.)

To evaluate the obtained effect sizes in this meta-analysis, Cohen's (1988) criteria were used. According to these criteria, the effect size could be measured in three levels: if effect size values are 0.20 or below, it is interpreted as a small effect; if between 0.20–0.80, it is moderate; if 0.80 and above, it is interpreted as a large effect. As stated by Borenstein, *et al.* (2009), there may be different effects underlying different studies. For example, in this meta-analysis, the effect size might be higher (or lower) in studies where a particular stage of writing is developed or the participants have a higher educational level, etc; therefore, the random-effects model is used. The choice of the random-effects model is because it is assumed that the true effect size varies from study to study, and the summary effect is the estimate of the mean of the distribution of effect sizes (Borenstein, *et al.*, 2009).

Data analysis

In this study, the author used the comprehensive meta-analysis (version 3) to statistically analyze the effect sizes. A comprehensive meta-analysis is a software that used solely for meta-analysis research. According to Bax *et al.* (2007), this software has the highest profile in the internet search engines of all included programs. It offers a comprehensive analysis of the effect sizes of different formats. The program features all major graphical presentations. It also presents graphs for publication bias. These features make it one of the best programs to be used in meta-analysis studies.

As the Q-statistic is used for multiple significance testing across several means, it was used to determine heterogeneity among the sampled study properties. Borenstein *et al.* (2009) indicated that the Q-statistic and p-value could be used for testing the null hypothesis but should not be used to estimate the true variance. For example, the significant low p-value ($p < 0.001$) does not indicate greater heterogeneity than the p-value of ($p < 0.049$); however, the statistically significant p-value ($p < 0.05$) indicates that heterogeneity exists (Lipsey and Wilson, 2001).

In addition to the Q-statistic and p-value, the I-square statistic can also be used to show that the variation is not due to chance, but rather indicates the heterogeneity of the sample; such heterogeneity exists only with high I-square values; a low I-square statistic is represented by a value of 25%, 50% represents a medium I-square statistic, whereas a high I-square statistic is represented by 75% (Higgins *et al.*, 2003).

Finally, 95% confidence intervals (CI) were calculated, based on procedures suggested by Lipsey and Wilson (2001), to test the statistical trustworthiness of the individual and averaged effect sizes. Confidence intervals are a good indicator for the trustworthiness of the effect sizes; if the confidence intervals include zero, it means that the true effect size is zero, and thus not trustworthy.

Results

The overall effect size of the eighteen articles will be discussed here. The analysis of these studies is made across various categories. First, the overall effect of technology on second/foreign language writing is discussed. Then this discussion will be followed by an analysis of the moderators that may affect second language (SL)/Foreign language (FL) writing learning.

The overall effect of the integration of technology in FL/SL writing

Table 1 shows the results obtained from the preliminary analysis of the 18 studies. It shows the overall effect of using technology in learning SL/FL Writing. Besides, the total number of participants in each group (experimental and control), and the standard deviation (SD) were given.

As shown in Table 1, technology had a large effect on second/foreign language writing; the overall effect size of using technology was large (d = 1.72). There was a statistically significant positive effect size, with a confidence interval greater than zero, ranging from 0.979 to 2.195. This suggested that TBW was more effective than non-TBW. In addition, the Q-value statistic showed that the overall sample was heterogeneous, Q (17), I-square = 94.8, $p < 0.05$. The I-square statistic indicated that the variability among the effect sizes was greater than that expected from a sampling error. Therefore, the results indicated that using technology for learning writing had a large effect size on learners' performance.

Analysis of moderators that may affect the integration of technology in SL/FL writing

The moderators that may affect the integration of technology in second/foreign language writing were analyzed according to certain categories for every moderator. The analyses of these moderators were explained below with all the necessary details.

The writing stages. The first moderator explored in this meta-analysis was the stages of writing. The process of writing took place in a number of stages as follows: prewriting, drafting, revising and editing. Based on the targeted stages in the eighteen studies, those studies were classified into four categories, namely, drafting, editing, drafting and editing and all stages. In total, 7 out of the 18 studies aimed to develop the drafting stage; one of these studies tried to improve the editing stage; six were conducted to develop both drafting and editing stages. Four studies aimed to improve all the four writing stages.

Table 2 presents the results obtained from the preliminary analysis of the above moderators. First, it shows that the average effect size of studies that aimed to develop the drafting stage was 1.66. This means that technology had a large effect size on the drafting stage. Additionally, there were positive confidence intervals, ranging from 0.574 to 2,493, which indicated that groups using TBW at the drafting stage performed better than non-TBW groups. Second, the effect size of the study that tried to improve the editing stage was 0.95, which indicated the large effect size of technology on the editing stage. Besides, the confidence intervals were positively ranging from 0.198 to 1.70, which assured that the group using TBW at the editing stage performed better than the non-TBW group.

Thirdly, the average of the effect size of studies that aimed to develop both drafting and editing was 1.25. This indicated that technology had a large effect size on developing the two stages, namely, drafting and editing together. Moreover, there were positive confidence

Table 1. Overall effect of integration of TBW in SL/FL writing

k*	Exp. N	Control N	Point estimate	SD Error	Confidence intervals Lower limit	Upper limit	p-value	Q-value	df (Q)	I-squared	d
18	560	721	1.587	0.310	0.979	2.195	0.000	330.514	17	94.856	1.7217

Note: k = number of studies

intervals, ranging from 0.664 to 1.634, which implied that groups using TBW at the two stages (drafting and editing) together performed better than non-TBW groups.

Finally, the average effect size of studies that aimed to improve all the four writing stages was 2.72, which implied a large effect size of technology on all the four stages, taken together. The confidence intervals were ranging from 0.448 to 5.779, indicating that groups using TBW at the four writing stages performed better than non-TBW groups.

The language context

The second moderator explored in this meta-analysis was the language context. The language context in which these studies were conducted was classified into two categories, namely, foreign language context and second language context. The results shown in Table 3 indicated that the use of technology in EFL writing classes had a large effect size (d = 2.026) and the confidence intervals were positive, ranging from 1.101 to 2.757, which indicated that foreign language learners who use TBW performed better than foreign language learners without TBW.

On the other hand, the results revealed that the use of technology in ESL writing classes had a small effect size (d = 0.195), but the confidence intervals were positive, ranging from −0.167 to 0.633, which clarified that second language learners who use TBW performed better than second language learners without TBW. Though the effect was small, there was still an effect of technology on second language learners. The small effect may be due to the high level of second language learners or to the point that we have only one study for second language learners in this meta-analysis. Most of the experimental studies found by the researcher were conducted in a foreign language context.

The learners' educational level

The third moderator was the learners' educational level. According to this moderator, the studies were classified into four groups, namely, high school, language institute, university and postgraduates. The 18 studies were distributed on these four levels as follows: four studies were conducted in high schools; three in language institutes; ten at universities; and one study took place after university (at the postgraduate level). The results shown in Table 4 indicated that the high school or university learners' use of technology had a large

Moderators	Categories	K	Confidence intervals Lower limit	Upper limit	p-value	Q-value	I-squared	d
Stages	Drafting	7	0.574	2.493	0.000	131.392	95.434	1.6604
	Editing	1	0.198	1.708	1.000	0.000	0.000	0.9532
	Drafting and editing	6	0.664	1.634	0.008	15.671	68.094	1.2538
	All stages	4	−0.448	5.779	0.000	177.928	98.314	2.7228

Table 2. Moderator analyses across contexts (stages of writing)

Moderators	Categories	K	Confidence intervals Lower limit	Upper limit	p-value	Q-value	I-squared	D
Context	Foreign Language	15	1.101	2.757	0.000	288.218	95.143	2.0269
	Second Language	3	−0.167	0.633	0.064	5.486	63.545	0.1955

Table 3. Moderator analyses across contexts (language context)

effect size (d = 2.305 and 2.129, respectively). However, the effect size of language institute learners' use of technology was small (d = 0.0499), and it was medium in case of the postgraduate learners (d = 0.3305).

Confidence intervals, as shown in Table 4, were positive in all the four cases. This implied that learners who used technology for learning second/foreign language writing at either high schools, language institutes, universities or at the postgraduate level performed better than learners who did not use technology for learning second/foreign language writing.

The learners' target language proficiency level

The fourth moderator was the learners' target language proficiency level. The proficiency levels used in this study were categorized into four groups, namely, beginner, intermediate, advanced and not found; the last category referred to the studies in which the language proficiency level of the subjects was not found. As shown in Table 5, the effect size of using technology in writing learning for beginners was medium (d = 0.50), whereas the effect sizes for intermediate and advanced learners were large (d = 1.1112 and 1.4529, respectively); the effect size of the studies in which the language proficiency level of the subjects was not found was also large (d = 2.8600).

Confidence intervals were positive in the cases of intermediate and advanced learners and with no language proficiency level studies, which indicated that learners using technology for learning second/foreign language writing at the intermediate and advanced levels performed better than learners who did not use computers for learning second/foreign language writing. On the other hand, the confidence intervals for beginner learners were negative, ranging from (0.241) to (0.619). This result could be attributed to the beginners' inability to use technological devices in developing their writing skills.

Discussion

The present study is designed to determine the effect of technology on EFL/ESL writing skills. Returning to the first question posed at the beginning of this study, it is possible to state that learners who use technological devices to learn second/foreign language writing

Table 4. Moderator analyses across contexts (learners' educational level)

Moderators	Categories	K	Confidence intervals Lower limit	Confidence intervals Upper limit	p-value	Q-value	I-squared	D
Educational level	High School	4	0.550	3.650	0.000	93.406	96.788	2.3051
	Language Institute	3	−1.509	1.595	0.000	27.530	92.735	0.0499
	University	10	1.090	3.000	0.000	172.035	94.769	2.1290
	Postgraduate	1	−0.163	0.824	1.000	0.000	0.000	0.3305

Table 5. Moderator analyses across contexts (the linguistic level)

Moderators	Categories	K	Confidence intervals Lower limit	Confidence intervals Upper limit	p-value	Q-value	I-squared	D
Language level	Beginner	2	0.241	0.619	0.961	0.002	0.000	0.5050
	Intermediate	5	−0.525	2.709	0.000	81.321	95.081	1.1112
	Advanced	5	0.149	2.603	0.000	69.911	94.278	1.4529
	Not found	6	1.187	4.312	0.000	149.985	96.666	2.8600

perform better than learners being taught in traditional ways. In spite of the differences in the effect sizes, which may be attributed to different factors, the results of this investigation also show that technology, in general, has a large effect on teaching second/foreign language writing. All the 18 studies, included in this meta-analysis, concluded with significant differences in the scores of learners who used technology (the experimental groups). The post-test scores in the selected studies were used to obtain the weighted effect sizes of these studies. Due to the absence of the delayed post-tests, the delayed post-test effect sizes were not calculated.

The second question of this study was to find out the relationship between technology-based writing (TBW) and the following moderators: stage of writing, language context, learners' target language proficiency and language level. In response to this question, the author conducted analyses for the effects of technology on writing learning with reference to these four moderators. These analyses are discussed below:

Stage of writing

To investigate whether the stage of writing influences the impact of technology on second/foreign writing learning, effect sizes were calculated in four categories, namely, drafting, editing, drafting and editing and all stages. The results of this meta-analysis revealed that technology could help second/foreign language learners to improve their writing regardless of the stage of writing being targeted. The effect sizes for studies targeting each of the four categories were large. However, it has also been shown that the effect size of studies that aimed to improve the four stages together was the largest. This may be due to the point that these studies tried to improve the learners' overall writing performance (Janfaza *et al.*, 2014). They also tried to develop the different sub-skills of writing (Ahmed, 2013) and to improve not only the learners' ability to write but also their ability to think of what to write and to edit the other learners' pieces of writing (Kashani *et al.*, 2013; Khalil, 2017).

Language context

Regarding the effect of language context on using technology in writing learning, the results of this meta-analysis indicated that technology had a greater effect among foreign language learners than what it had with second language learners. Large effect sizes were reported for most of the studies investigating the effect of technology on writing learning in foreign language contexts, with the exception of some studies (Estarki and Bazyar, 2016; Hsu and Lo, 2018; Kashani *et al.*, 2013; Khaksefidi, 2016), who reported medium effect sizes. On the other hand, the effect sizes of studies with the same aim in second language contexts showed a small effect size (Kimmons *et al.*, 2017; Lam *et al.*, 2018). Only 2 of the 18 studies had negative effect sizes, one in the foreign language context (Janfaza *et al.*, 2014) and another in the second language context (Bikowski and Vithanage, 2016).

The learners' educational level

To explore the impact of the learners'educational level on technology-based writing, the effect sizes of four categories of the educational level were calculated. These categories are high school, language institute, university and postgraduate. The results of this meta-analysis showed large effect sizes of technology on writing learning at high school and university levels, whereas small effect sizes were reported at language institutes and postgraduate levels. A possible explanation for these results may be the lack of adequate studies at the language institutes and postgraduate levels: only three studies were conducted for language institutes and one for postgraduate. Therefore, it is difficult to generalize this finding.

The learners' target language proficiency level

The results of the effects of learners' target language proficiency level on technology-based writing revealed a greater effect among studies in which the learners' levels were not mentioned. The effect sizes were calculated in four levels, namely, beginner, intermediate, advanced and not found. The fourth level was specified for studies in which the learners'' proficiency level was not identified. The results showed large effect sizes in all the levels except the first level (beginner), which showed a medium effect size. This result can be attributed to the beginners' limited abilities to use technological devices for learning purposes. Additionally, learners at the intermediate and advanced levels are more exposed to the language than learners at the beginner level.

Conclusion

The main goal of the current study was to use a meta-analysis design to determine the overall effectiveness of technology in learning second/foreign language writing. This study has generally shown that technology has a positive effect in improving the learning of second/foreign language writing. Regarding the moderators that have been considered in this meta-analysis, the study has shown that most researchers concentrate on using technology in developing two stages, namely, drafting and editing. In spite of its importance, the first writing stage, that is prewriting, has not received enough consideration. The results of this investigation have also revealed that studies investigating the effect of technology on writing learning in the foreign language context are more in number than studies with the same goal in the second language context. Most of the studies included in this meta-analysis were conducted in foreign language situations. It has also been noted that technology has a greater effect on the learners' writings at high school and university levels than it has on these learners at the language institutes and postgraduate levels. It has also been noticed that most of the previous studies were conducted for intermediate and advanced learners. Few studies were directed to beginner learners. Besides, the smallest effect of technology was on learning writing for beginner learners.

Implications

The findings of this study suggest that technology is an effective means which can be used in improving second/foreign language writing. These findings have a number of important implications for future practice. The first implication is that teachers of writing are strongly recommended to integrate technology effectively with their methods of teaching. Integrating technology into teaching methodologies will not only increase the learners' motivation but also promote the teaching process, in general (Karami *et al.*, 2019; Wihastyanang *et al.*, 2020). Therefore, instructors of writing have to design activities in which learners can get benefits from their passion for technology in improving their abilities to write. These activities should not only be confined to classrooms. They should include some tasks that require learners to use technology outside classrooms.

The second implication of this meta-analysis is that it attracts the learners' attention to the importance of technology in the development of their writing proficiency. Such importance is promoted because of Coronavirus. This world pandemic makes it clear that traditional learning tools will not be effective in the post-Corona world. This meta-analysis helps learners recognize the most useful technological devices they can use in developing their abilities to write.

Another important practical implication is that course designers can get benefits from the modern technological devices used in the references of this meta-analysis. Newer technologies, more modern technological devices, and more recent computer and mobile applications are used in language learning daily. So course designers are suggested to include these technologies while designing or developing the courses of writing. Examples of the new technologies that have been revealed in the area of teaching writing are electronic portfolio that was used in Karami *et al.* (2019) and Barrot (2020), digital sandbox gaming which was suggested by Ellison and Drew (2020), the flipped classroom which was followed by Alghasab (2020) and Google docs as used by Alharbi (2019).

Limitations and future directions

Several limitations can be identified in this meta-analysis. First, the publication bias may not have been controlled sufficiently in this study owing to the limited number of papers published and because only peer-reviewed studies were included. Articles published in journals tend to report a positive impact of technology on writing skills. Second, the number of primary studies included in this analysis is very limited. This is because of the limited number of experimental studies, in general, and the limited number of experimental studies that have clear data regarding the means and standard deviations, in particular. Due to their unclear data, a lot of experimental studies have been excluded. Future meta-analyses are suggested to have more studies to make them more generalizable. The third limitation lies in the point that all the studies included in this meta-analysis dealt with the effect of technology on one or two sages of the writing process, i.e. drafting and editing. Due to its importance, further studies should investigate the effect of technology on the prewriting and revising stages. The fourth limitation of this meta-analysis is in terms of language contexts. All the studies included in this meta-analysis were taken from two contexts, namely, foreign language and second language contexts. The effect of technology on the first language writing performance has not been measured. This issue can be tackled in future research.

All the studies included in this meta-analysis dealt with the writing skills of the adult learners, at least high school leaners. Further researchers are required to investigate the effect of technology on younger language learners. As technology is now within the children's hands, a study can be conducted to investigate, for example, its effect on the children's ability to write a simple paragraph. This meta-analysis also deals with the effect of technology on the learners' writing performance, in general, without discussing the differences between the effect of computer and mobile applications. In spite of the fact that the moderator of the technological devices is an important factor, it has not been included as one of the moderators in this meta-analysis. This is because, in many of the selected studies, the technological devices used, such as Twitter, email and WhatsApp, can be accessed by both a computer and a mobile. Another moderator that was not considered in this meta-analysis is gender. The current study did not consider the gender of the participants. Further research can take into account the differences in the effect of technology on the performance of male learners and that of female ones.

To conclude, it is important to consider all the above limitations if we want to have a complete picture of the effect of technology on second/foreign language writing.

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Appendix 1

Category	Subcategories	*N*
Publication	Journal	18
Stages of writing	Drafting	7
	Editing	1
	Drafting and editing	6
	All Stages	4
Language context	Foreign Language	17
	Second Language	1
Educational levels	School	4
	Language Institutes University	3
		10
	Postgraduate	1
Language proficiency	Beginner	2
	Intermediate	5
	Advanced	5
	NA	6

Table A1.
Coding scheme of the studies included in the meta-analysis

Appendix 2

Table A2. Journals and other details for the selected studies in this meta-analysis

No.	Author's name	Journal	Age	Technological device	Stage	Context	Educational level	Language level
1	Abdul Fattah (2015)	*Journal of Education and Practice*	NF	WhatsApp	Drafting and editing	FL	University	Advanced
2	Ahmed (2015)	*AWEG*	NF	Twitter	Drafting	FL	University	NF
3	Ahmed (2013)	*AWEG*	NF	E-mail	All stages	FL	High School	NF
4	Bikowski and Vithanage (2016)	*Language Learning and Technology*	18–27	Web-based	Drafting	SL	University	Advanced
5	Ciftci and Kocoglu (2012)	*Educational Computing Research*	20–24	Blogs	Editing	FL	University	NF
6	El-Ghonaimy (2015)	*JRCIET*	NF	Computer	Drafting and editing	FL	University	NF
7	Estarki and Bazyar (2016)	*European Online Journal of Natural and Social Sciences*	15–21	Mobile	Drafting	FL	Institute	intermediate
8	Ghafoori *et al.* (2016)	*IJLLALW*	20–25	Computer	Drafting	FL	University	intermediate
9	Hsu and Lo (2018)	*Language Learning and Technology*	20–21	Wiki	Drafting	FL	University	Beginner
10	Jafarian *et al.* (2012)	*European Journal of Social Sciences*	17–19	A software computer program	Drafting	FL	High School	Advanced
11	Janfaza *et al.* (2014)	*Advances in Language and Literary Studies*	16–18	E-mail	All stages	FL	Institute	intermediate
12	Kashani *et al.* (2013)	*English Language Teaching*	NF	Blogs	All stages	FL	Postgraduate	Advanced
13	Khaksefidi (2016)	*Journal of Language Sciences and Linguistics*	NF	Mobile	Drafting and editing	FL	University	NF
14	Khalil (2017)	*JRCIET*	NF	WhatsApp	All stages	FL	University	Advanced
15	Khoshsima and Sayadi (2016)	*Advances in Language and Literary Studies*	18–22	Virtual learning	Drafting and editing	FL	University	intermediate
16	Kimmons *et al.* (2017)	*Computers and Composition*	13–14	Chromebook	Drafting	SL	High School	Beginner
17	Lam *et al.* (2018)	*Language Learning and Technology*	16–17	Blended learning	Drafting and editing	SL	High School	NF
18	Malekzadeh and Najmi (2015)	*Journal of Applied Linguistics and Language Research*	16–23	Mobile	Drafting and editing	FL	Institute	Intermediate

Appendix 3

Table A3. Individual effect sizes for studies about technology and writing

Study	Experimental group *N*	M	SD	Control *N*	Mean	SD	*D*
Abdul Fattah (2015)	15	13.93	3.99	15	8.8	3.75	1.3249
Ahmed (2015)	30	20	1.84	30	11.3	1	5.8752
Ahmed (2013)	30	75.83	5.58	30	28.87	7.18	7.3033
Bikowski and Vithanage (2016)	32	84.09	9.28	27	86.51	11.84	−0.2299
Ciftci and Kocoglu (2012)	15	72.82	4.3	15	66.87	7.71	0.9532
El-Ghonaimy (2015)	19	21.03	4.22	19	14.21	2.99	1.8649
Estarki and Bazyar (2016)	15	18.27	2.37	15	16.93	1.83	0.6321
Ghafoori *et al.* (2016)	25	16.28	1.41	25	12	1.23	3.2349
Hsu and Lo (2018)	26	4.9	1.09	26	4.15	1.46	0.5821
Jafarian *et al.* (2012)	20	80	13.35	20	63.75	16.05	1.1008
Janfaza *et al.* (2014)	21	17.38	3.69	21	22.55	3.28	−1.4809
Kashani *et al.* (2013)	33	65.48	7.291	31	63.11	7.042	0.3305
Khaksefidi (2016)	60	14.23	2.27	60	12.08	3.2	0.775
Khalil (2017)	25	16.56	2.022	23	5.43	2.66	4.7385
Khoshsima and Sayadi (2016)	10	12.75	1.419	10	9.8	1.295	2.1714
Kimmons *et al.* (2017)	139	6.59	2.16	319	5.73	1.94	0.428
Lam *et al.* (2018)	30	2.13	0.43	20	1.95	0.51	0.3885
Malekzadeh and Najmi (2015)	15	17.27	1.58	15	15.6	1.76	0.9986

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